

Introduction to CMP2020

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CMP2020 Artificial Intelligence

Lecturer

Helen Harman (she/her)

- Programme leader for CS
- Research areas:
  - Applied AI
  - Task planning and allocation
  - Robotics

Office hours: Thursdays 11am





Module Learning Outcomes

[LO1] Demonstrate an understanding of the theoretical capabilities of AI

[LO2] Successfully apply Artificial Intelligence techniques to practical problems

[LO3] Demonstrate an ability to identify relevant ethical and responsible AI issues

Module Overview

	Workshop	Lecture
1	*No workshop*	Introductions
2	Types of Agent	Search
3	Search	Responsible AI
4	Responsible AI	Heuristic Search
5	Heuristic Search	Optimisation / GAs
6	Genetic Algorithms	Adversarial Search
-	-	-

	Workshop	Lecture
8	Adversarial Search	Logical Agents
9	Basics of prolog	First Order Logic
10	Prolog	Classical planning
11	Assessment 2 help	Planning in the real world
12	PDDL	Learning to do
-	-	-
16	Learning to do	Wrap-up

Lectures: 4pm-5pm Fridays
Workshops: 11am-12pm or 12pm-1pm Fridays

Textbook

Artificial Intelligence: A Modern Approach
By Stuart Russell and Peter Norvig

3rd (2010) or 4th Edition (2021)

Henceforth "R&N"

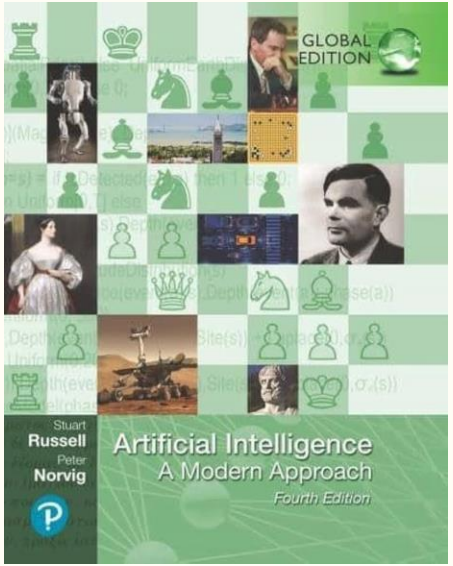


Image taken from Amazon

Assessments

Assessment 1 (40%): Presentation / video

Assessment 2 (60%): Software & short report

Check them out on Blackboard
(… pending moderation)

Builds on your knowledge from 1st year.



Introduction to AI

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CMP2020 Artificial Intelligence

Contents

1. **What is AI?**
2. Intelligent agents
3. Some simple agents
4. Environments

What is AI?

Enter your thoughts here:



pollEv.com/HelenHarman

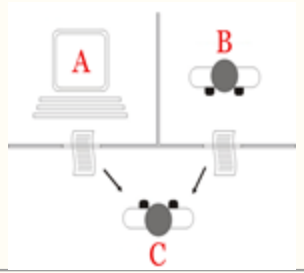
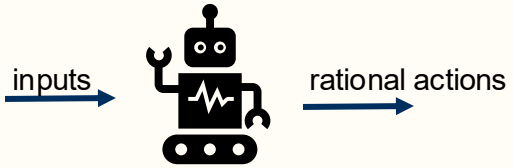
Discuss your thoughts with your neighbour.

What is AI?

AI is both science and engineering:

- The *science* of understanding intelligent agents – explain and predict behaviors of entities
- The *engineering* of intelligent entities

Four views of AI

Thinking Humanly: Cognitive modeling. "Machines with minds" 	**Thinking Rationally:** logical approaches “Socrates is a man; all men are mortal; therefore, Socrates is mortal."
Acting Humanly: The Turing Test 	**Acting Rationally:** rational agents 

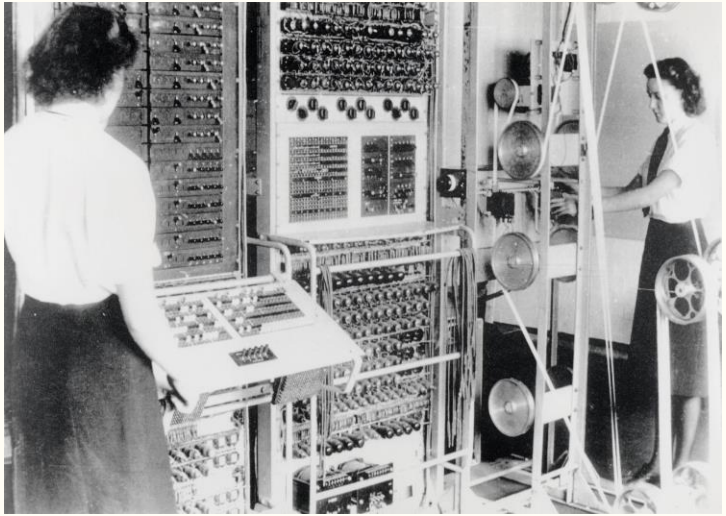
Acting Humanly

- When can we consider a machine to be intelligent?
- Alan Turing
- "Imitation Game" 1950



(picture of Alan Turing from 1951)

~75% of the staff at Bletchley park were women.



(Dorothy Du Boisson and Elsie Booker at Bletchley Park)

Thinking Humanly

Understanding how the mind works — how do we think?

Three possible routes:

- introspection — we figure it out ourselves!
- experiment — draw upon techniques of psychology to conduct controlled experiments.
- observation — draw upon neuroscience to figure out the structure of the brain, then try to replicate it.

The discipline of cognitive science: still influential in areas like human-robot interaction, explainability and some research into vision and natural language processing.

Thinking Rationally

How should we think?

- Use logic to capture the laws of rational thought as symbols.
- Reasoning involves shifting symbols according to well-defined rules (like algebra).
- Result is idealised reasoning.
- Logicist approach is theoretically attractive.

Acting Rationally

Acting to achieve one's goals, given one's beliefs.

- An agent is a system that perceives and acts; intelligent agent is one that acts rationally w.r.t. the goals we delegate to it.
- Emphasis shifts from designing theoretically best decision-making procedure to best decision-making procedure possible in circumstances.
- Logic may be used in the service of finding the best action — not an end in itself
- Achieving perfect rationality — making the best decision theoretically possible — is not usually possible, due to limited resources (time, computational power, memory, uncertainty).

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Intelligent agents



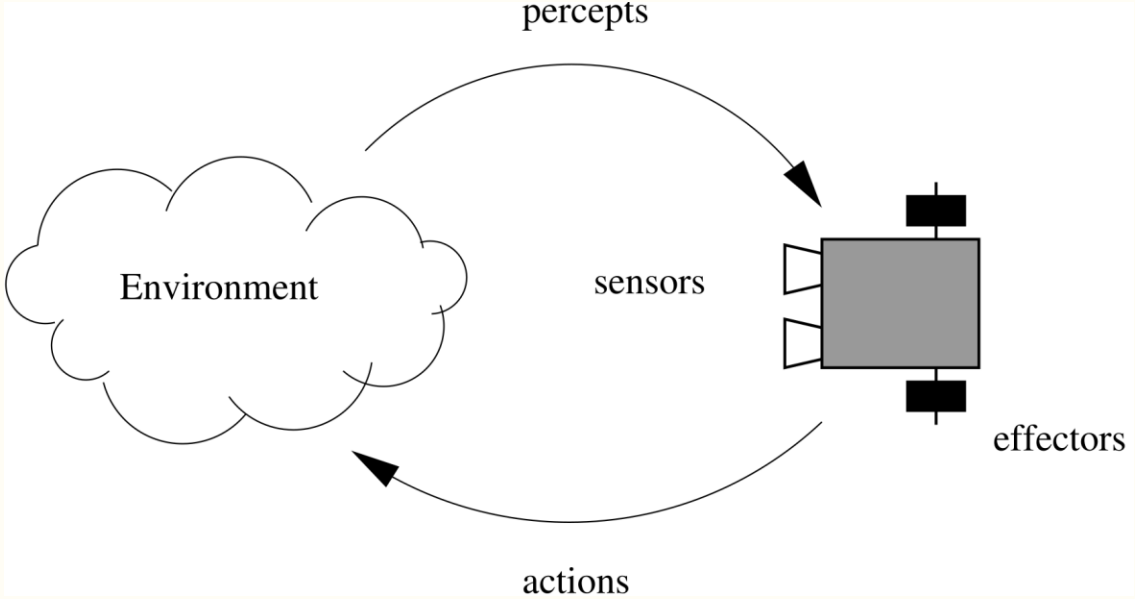
Good enough to be useful.

An agent is a system that:

- is situated in an environment,
- is capable of perceiving its environment, and
- is capable of acting in its environment

with the goal of satisfying its design objectives.

Intelligent agents



Considering the environment, the task is to program the agent to convert percepts to actions.

Human agents



What environment: physical world
What sensors: eyes, ears,...
What effectors: hands, legs...

Bot



What environment?
What sensors?
What effectors?

Bot



What environment: Unreal
What sensors: List of objects and locations
What effectors: move, shoot, change weapon

Robot



What environment: physical world
What sensors: sonar, camera ...
What effectors: wheels, legs...

What to do?

Key problem: knowing the right thing to do.

Easy in principle: Consider all alternatives and choose the "best".

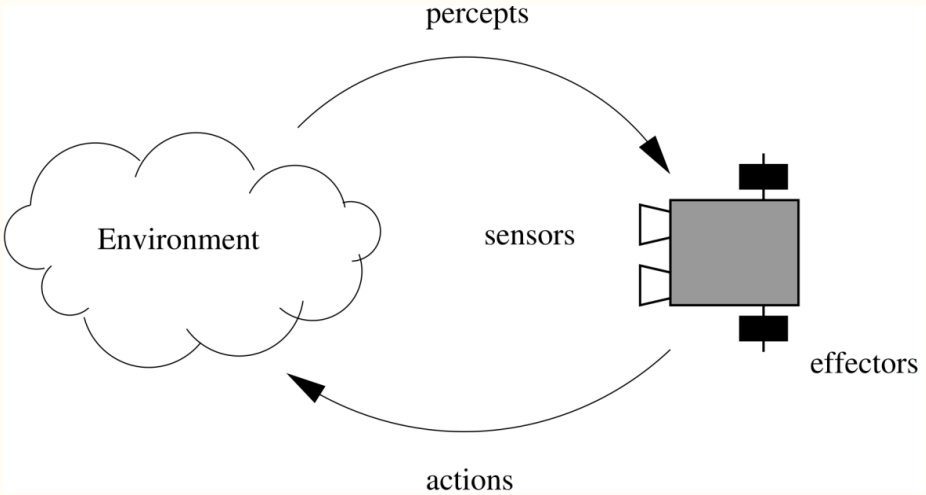
But: We need to decide in time for that decision to be useful.

A tradeoff.

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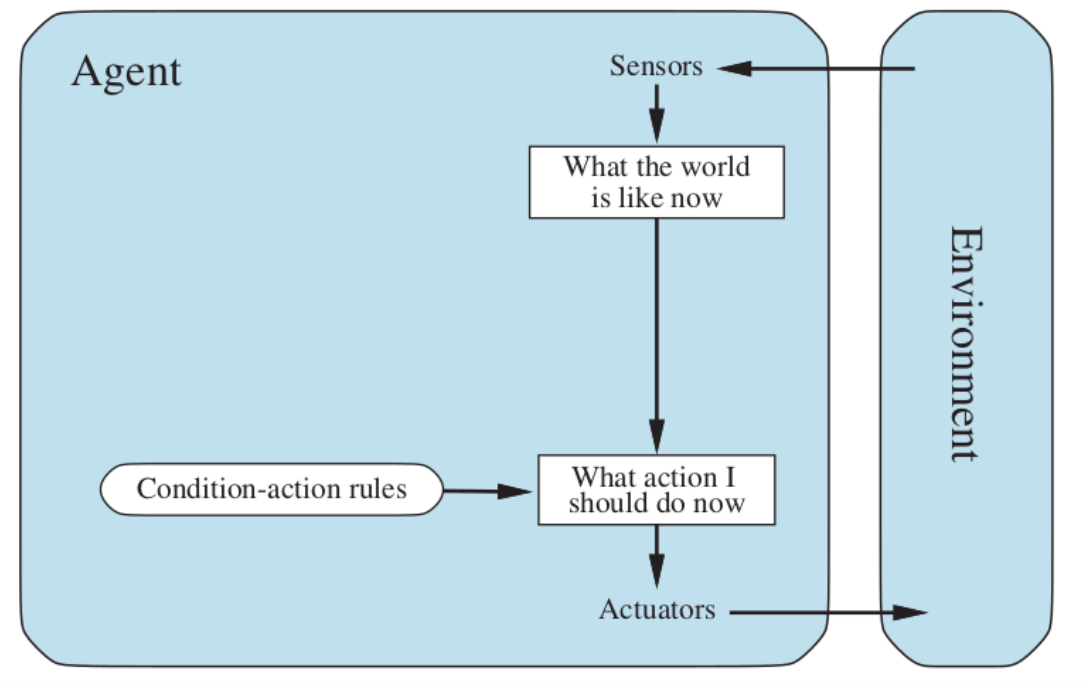
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Simple agents



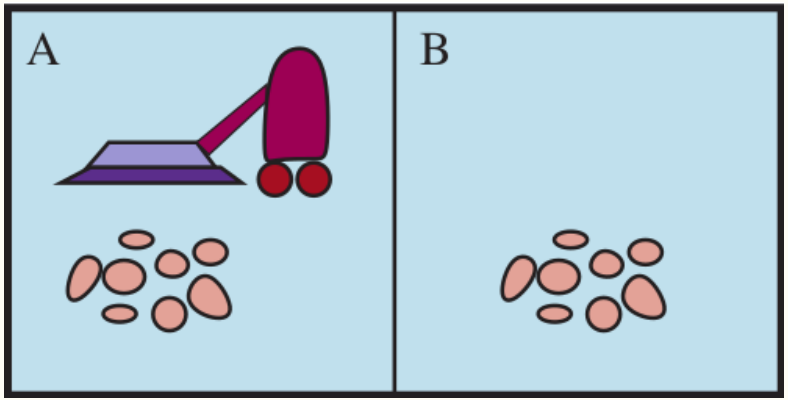
Every agent maps percepts to actions

Simple reflex agent



Simple reflex agent

In the "vacuum world":





a simple agent may suffice.

Simple reflex agent

```
function Reflex-Vacuum-Agent( [location, status]) returns an action
        if status = Dirty then return Suck
        else if location = A then return Right
        else if location = B then return Left
```

A more general version

function Simple-Reflex-Agent(percept) **returns** an action
    Static: rules, a set of condition-actions rules

    state ← Interpret-Input(percept)
    rule ← Rule-match(state, rules)
    action ← rule.action
    return action

More on different types of agents in your workshop

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Fully observable vs partially observable

- Fully observable environment (aka accessible): the agent can obtain complete, accurate, up-to-date information about the environment's state.
- Most moderately complex environments are only partially observable (aka non-accessible).





The more observable an environment is, the simpler it is to build agents to operate in it

Deterministic vs non-deterministic

- A deterministic environment is one in which any action has a single guaranteed effect — there is no uncertainty about the state that will result from performing an action.
- The physical world can to all intents and purposes be regarded as non-deterministic.
- R & N calls non-deterministic environments stochastic if we quantify the non-determinism using probability theory.
- Non-deterministic environments present greater problems for the agent designer.

Episodic vs sequential

- Episodic environment: the performance of an agent is dependent on a number of discrete episodes, with no link between the performance of an agent in different scenarios.
  - Assembly line where an agent had to spot defective parts.

- Episodic environments are simpler from the agent developer's perspective because the agent can decide what action to perform based only on the current episode — it need not reason about the interactions between this and future episodes.

- Non-episodic or sequential: the current decision affects future decisions.
  - Driving a car is sequential.

Static vs dynamic

- A static environment is one that can be assumed to remain unchanged except by the performance of actions by the agent.
- If the environment changes while the agent is deliberating, then it is dynamic.
- A dynamic environment is one that has other processes operating on it, and which hence changes in ways beyond the agent's control.
- The physical world is a highly dynamic environment.
- One reason an environment may be dynamic is the presence of other agents

Discrete vs continuous

- An environment is discrete if there are a fixed, finite number of actions and percepts in it.
- R & N gives a chess game as an example of a discrete environment, and taxi driving as an example of a continuous one.





How would you classify the environment of a Poker (card game) playing agent?

Example: Poker

- Partially observable
- Non-deterministic (stochastic)
- Sequential
- Static (environment does change whilst agent is deliberating)
- Discrete

Summary

This lecture has looked at:

- Definitions of artificial intelligence
- An introduction to intelligent agents
- Types of environment

Broadly speaking, the rest of the module will dig into more sophisticated ways of deciding what an agent should do.

Workshop: more types of agents – in Python